Cambridge International AS & A Level

Cambridge Assessment International Education
Cambridge International Advanced Subsidiary and Advanced Level

MATHEMATICS **9709/63**

Paper 6 **May/June 2019**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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[Turn over

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

- The symbol FT implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.

  - Note: B2 or A2 means that the candidate can earn 2 or 0.
    B2/1/0 means that the candidate can earn anything from 0 to 2.

  The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.

- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF/OE Any Equivalent Form (of answer is equally acceptable) / Or Equivalent

AG Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)

CAO Correct Answer Only (emphasising that no “follow through” from a previous error is allowed)

CWO Correct Working Only – often written by a ‘fortuitous’ answer

ISW Ignore Subsequent Working

SOI Seen or implied

SR Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR –1 A penalty of MR –1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become “follow through” marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR –2 penalty may be applied in particular cases if agreed at the coordination meeting.

PA –1 This is deducted from A or B marks in the case of premature approximation. The PA –1 penalty is usually discussed at the meeting.

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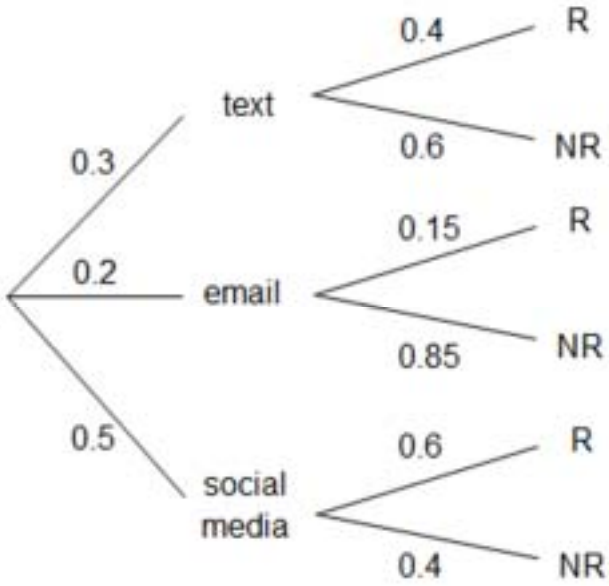
Question	Answer	Marks	Guidance
1(i)	$P(79 < X < 91) = P\left(\frac{79-85}{6.8} < Z < \frac{91-85}{6.8}\right)$ $= P(-0.8824 < Z < 0.8824)$	**M1**	Using ± standardisation formula for either 79 or 91, no continuity correction
	$= \Phi(0.8824) - \Phi(-0.8824)$ $= 0.8111 - (1 - 0.8111)$	**M1**	Correct area ($\Phi - \Phi$) with one +ve and one –ve z-value or $2\Phi - 1$ or $2(\Phi - 0.5)$
	$= 0.622$	**A1**	Correct answer
		3	
1(ii)	$z = -1.751$	**B1**	± 1.751 seen
	$-1.751 = \frac{t-85}{6.8}$	**M1**	An equation using ± standardisation formula with a z-value, condone σ^2 or $\sqrt{\sigma}$
	$t = 73.1$	**A1**	Correct answer
		3	

Question	Answer	Marks	Guidance
2(i)		**B1**	Fully correct labelled tree with correct probabilities for 'Send'
		B1	Fully correct labelled branches with correct probabilities for the 'reply'
		2	

Question	Answer	Marks	Guidance
2(ii)	$\mathrm{P}(email \mid NR) = \dfrac{\mathrm{P}(\mathrm{email} \cap \mathrm{NR})}{\mathrm{P}(\mathrm{NR})} = \dfrac{0.2 \times 0.85}{0.3 \times 0.6 + 0.2 \times 0.85 + 0.5 \times 0.4}$	**M1**	P(email) × P(NR) seen as numerator of a fraction, consistent with *their* tree diagram
	$= \dfrac{0.17}{0.18 + 0.17 + 0.2} = \dfrac{0.17}{0.55}$	**M1**	Summing three appropriate 2-factor probabilities, consistent with *their* tree diagram, seen anywhere 0.55 oe (can be unsimplified) seen as denom of a fraction
	$= 0.309,\ \dfrac{17}{55}$	**A1**	
		A1	Correct answer
		4	

Question	Answer	Marks	Guidance
3(i)	9! × 2	**B1**	9! seen multiplied by $k \geqslant 1$, no addition
	= 725760	**B1**	Exact value
		2	
3(ii)	Eg ($K_1K_2K_3K_4K_5$) A A A (U_1U_2) A	**B1**	2! or 5! seen mult by k > 1, no addition (arranging Us or Ks)
	= 5! × 2! × 6!	**B1**	6! Seen mult by k > 1, no addition (arranging AAAAKU)
	= 172800	**B1**	Exact value
		3	

Question	Answer	Marks	Guidance
4(i)	M(8) W(4) 4 2 in ${}^8C_4 \times {}^4C_2 = 420$ ways 5 1 in ${}^8C_5 \times {}^4C_1 = 224$ ways 6 0 in ${}^8C_6 \times {}^4C_0 = 28$ ways	**B1**	One unsimplified product correct
		M1	Summing the number of ways for 2 or 3 correct scenarios (can be unsimplified), no incorrect scenarios
	Total 672 ways	**A1**	Correct answer
		3	

Question	Answer	Marks	Guidance
4(ii)	Total number of selections = ${}^{12}C_6$ = 924 (A)	**M1**	${}^{12}C_x$ – (subtraction seen), accept unsimplified
	Selections with males together = ${}^{10}C_4$ = 210 (B)	**A1**	Correct unsimplified expression
	Total = (A) – (B) = 714	**A1**	Correct answer
	Alternative method for question 4(ii)		
	No males + Only male 1 + Only male 2 = ${}^{10}C_6 + {}^{10}C_5 + {}^{10}C_5$	**M1**	${}^{10}C_x + 2 \times {}^{10}C_y$, $x \neq y$ seen, accept unsimplified
	= 210 + 252 + 252	**A1**	Correct unsimplified expression
	= 714	**A1**	Correct answer
	Alternative method for question 4(ii)		
	Pool without male 1 + Pool without male 2 – Pool without either male	**M1**	$2 \times {}^{11}C_x - {}^{10}C_x$
	= ${}^{11}C_6 + {}^{11}C_6 - {}^{10}C_6$ = 462 + 462 – 210	**A1**	Correct unsimplified expression
	= 714	**A1**	Correct answer
		3	

Question	Answer	Marks	Guidance
5(i)	$P(0, 1, 2) = (0.66)^{14} + {}^{14}C_1(0.34)(0.66)^{13} + {}^{14}C_2(0.34)^2(0.66)^{12}$	**M1**	Binomial term of form ${}^{14}C_x p^x(1-p)^{14-x}$ $0 < p < 1$ any p, $x \neq 14,0$
	$= 0.0029758 + 0.02146239 + 0.071866$	**A1**	Correct unsimplified answer
	$= 0.0963$	**A1**	Correct answer
		3	
5(ii)	Mean $=600 \times 0.34 = 204$, Var $= 600 \times 0.34 \times 0.66 = 134.64$	**B1**	Correct unsimplified *np* and *npq* (or sd = 11.603 or Variance = 3366/25)
	$P(< 190) = P\left(z < \frac{189.5 - 204}{\sqrt{134.64}}\right) = P(z < -1.2496)$	**M1**	Substituting *their* μ and σ, (no σ^2 or $\sqrt{\sigma}$) into the Standardisation Formula with a numerical value for *'189.5'*. Condone ± standardisation formula
		M1	Using continuity correction 189.5 or 190.5 within a Standardisation formula
	$= 1 - \Phi(1.2496)$	**M1**	Appropriate area Φ from standardisation formula P(z<….) in final solution, (<0.5 if z is –ve, >0.5 if z is +ve)
	$= 1 - 0.8944 = 0.106$	**A1**	Correct final answer
		5	

Question	Answer	Marks	Guidance
6(i)	<table><tr><td>score</td><td>1</td><td>2</td><td>3</td><td>4</td><td>6</td><td>9</td></tr><tr><td>prob</td><td>$\frac{3}{15}$</td><td>$\frac{4}{15}$</td><td>$\frac{4}{15}$</td><td>$\frac{1}{15}$</td><td>$\frac{2}{15}$</td><td>$\frac{1}{15}$</td></tr></table>	**B1**	Probability distribution table with correct scores, allow extra score values if probability of zero stated
		B1	2 probabilities (with correct score) correct
		B1	3 or more correct probabilities with correct scores
		B1	**FT** $\Sigma p = 1$, at least 4 probabilities
		4	
6(ii)	mean = $\frac{(3+8+12+4+12+9)}{15} = \frac{48}{15}$ (3.2)	**B1**	
	Var = $\frac{(3+16+36+16+72+81)}{15} - (\textit{their}\, 3.2)^2$	**M1**	**FT** Substitute *their* attempts at scores in correct var formula, must have "– mean^2" (condone probabilities not summing to 1)
	$= \frac{224}{15} - 3.2^2 = 4.69 \left(\frac{352}{75}\right)$	**A1**	
		3	
6(iii)	Score of 4, 6, 9	**M1**	Identifying relevant scores from *their* mean and *their* table
	Prob $\frac{4}{15}$ (0.267)	**A1**	Correct answer **SC** B1 for 4/15 with no working
		2	

Question	Answer	Marks	Guidance
7(i)	Thaters School | | Whitefay Park School 8 | 3 | 8 3 | 4 | 5 7 8 8 7 6 4 2 | 5 | 3 6 6 6 2 1 | 6 | 1 4 6 9 5 | 7 | 3 5 8 | 8 | 3 Key 8 | 4 | 5 represents 48 minutes for Thaters School and 45 minutes for Whitefay Park School.	**B1**	Correct stem can be upside down, ignore extra values,
		B1	Correct Thaters School labelled on left, leaves in order from right to left and lined up vertically, no commas
		B1	Correct Whitefay Park School labelled on same diagram on right hand side in order from left to right and lined up vertically, no commas
		B1	**FT** Correct key for *their* diagram, need both teams identified and 'minutes' stated at least once here or in leaf headings or title. **SC** If 2 separate diagrams drawn, **SCB1** if both keys meet these criteria
		4	
7(ii)	LQ = 50 UQ = 61.5	**B1**	Both quartiles correct
	IQ range = 61.5 – 50 = 11.5	**B1**	**FT** $61 \leqslant UQ \leqslant 62 - 48 \leqslant LQ \leqslant 52$
		2	
7(iii)	$\Sigma(x - 60)^2 = (-15)^2 + (-13)^2 + (-7)^2 + (-4)^2 + (-4)^2 + 1^2 + 4^2 + 6^2 + 9^2 + 13^2 + 23^2 + 15^2 + 18^2$	**M1**	Summing squares with at least 5 correct unsimplified terms
	= 1856	**A1**	Exact value
		2	

Question	Answer	Marks	Guidance
7(iv)	Var = mean of coded squares – (coded mean)^2 $= \frac{\Sigma(x-60)^2}{13} - \left(\frac{\Sigma(x-60)}{13}\right)^2$	**M1**	Using two coded values in correct formula (variance or sd)
	Var $= \frac{\textit{their}\,1856}{13} - \left(\frac{46}{13}\right)^2$ $= 130$	**A1**	Correct answer **SC** if correct variance obtained by another method give **SCB1**
		2	